# Sajini Tharushika

Software Engineering Undergraduate

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**GitHub:** github.com/Sajini3655 | **Portfolio:** sajini3655.github.io/portfolio/

## Profile

Information Technology undergraduate at the University of Moratuwa seeking a Software Engineering internship. Hands-on experience developing full-stack applications, REST APIs, and distributed backend systems using Java, Spring Boot, React, and PostgreSQL. Experienced in asynchronous processing, API management, authentication, and automated testing, with a strong foundation in object-oriented programming and problem-solving.

## Education

**University of Moratuwa, Sri Lanka** 2023 – 2027

B.Sc. (Hons) in Information Technology **Current GPA: 3.4**

**Sujatha Vidyalaya, Matara** G.C.E. Advanced Level – Mathematics Stream

Results: AAB | Z-Score: 1.87

## Projects

**FlowForge – API Management & Distributed Workflow Platform** Individual Project

- Developed a distributed workflow platform integrating WSO2 API Manager 4.7 with a Spring Boot backend for API lifecycle management and reliable asynchronous job execution.
- Implemented transactional outbox publishing, SHA-256-based request idempotency, RabbitMQ retry and dead-letter queues, and Redis distributed locking to prevent duplicate execution and handle processing failures.
- Integrated JWT-based authentication, Prometheus/Grafana monitoring, and Docker-based infrastructure with Kubernetes manifests. Added automated unit and integration tests using JUnit 5 and Testcontainers.
- **Tech:** Java, Spring Boot, PostgreSQL, WSO2 API Manager, RabbitMQ, Redis, Docker, Kubernetes, Prometheus, Grafana, GitHub Actions
  **GitHub:** github.com/Sajini3655/Flowforge

**Nexora – AI-Enhanced Project Management System** Group Project

- Co-developed a role-based project management platform for administrators, managers, developers, and clients, supporting project tracking, task assignment, ticket management, and team communication.
- **My Contribution:** Developed administrator and manager dashboards for user registration, role assignment, project management, and progress monitoring. Implemented an AI chat interface for conversation summarization and summary-to-ticket creation.
- **Tech:** Java, Spring Boot, Spring Security, React, PostgreSQL, Python, FastAPI, Groq API, RabbitMQ, WebSockets/STOMP, JWT, Docker
  **GitHub:** github.com/Sajini3655/Nexora
  **Live Demo:** nexora-three-peach.vercel.app/login

**PhysicsLab – Bilingual A/L Physics Learning Platform** Group Project

- Co-developed an English/Sinhala Physics learning platform featuring interactive virtual laboratories, guided lessons, physics simulations, and assessments for Sri Lankan A/L students.
- **My Contribution:** Implemented interactive physics practicals, calculation validation, and browser-local experiment persistence. Verified functionality with automated Vitest and Playwright tests.
- **Tech:** Next.js, React, TypeScript, Tailwind CSS, Vitest, Playwright
  **GitHub:** github.com/vishiiboy/physicslab
  **Live Demo:** physicslab-eta.vercel.app/en

**FabricLab – Interactive Material & Physics Laboratory** Individual Project

- Developed a browser-based cloth physics simulator using WebGPU compute shaders and Extended Position-Based Dynamics (XPBD), supporting real-time deformation, collision handling, and interactive 3D manipulation.
- Implemented configurable material presets, wind and gravity controls, deformation visualization, and experiment persistence. Built a 5K–100K-particle benchmarking suite with frame-time statistics and CSV export.
- **Tech:** TypeScript, WebGPU, WGSL, XPBD, GPU Compute Shaders, WebGL/Graphics Mathematics, Webpack, Node.js Test Runner, GitHub Actions, GitHub Pages
  **GitHub:** github.com/Sajini3655/fabriclab
  **Live Demo:** sajini3655.github.io/fabriclab/

**AURA – Full-Stack Fashion E-Commerce Web Application** Individual Project

- Built a full-stack fashion e-commerce platform with product browsing, shopping cart, checkout, order management, and a rewards system supporting vouchers and spin-the-wheel incentives.
- Developed backend API routes and database operations using Prisma and MongoDB, implementing JWT-based authentication, bcrypt password hashing, and cookie-based session handling.
- **Tech:** Next.js 15, React, JavaScript, Tailwind CSS, Prisma, MongoDB, JWT, bcrypt
  **GitHub:** github.com/Sajini3655/Aura

**LaserTunes – Laser-Based Musical Instrument** Group Project

- Co-developed a 12-laser interactive musical instrument integrating Arduino, ESP32, sensors, and audio hardware, supporting instrument switching, percussion effects, and loop recording. Presented at **FIT EXPO 2025**.
- **Tech:** Arduino Mega, ESP32, C++, VS1053B, Piezo Sensors, Embedded Systems
  **GitHub:** github.com/Sajini3655/LaserTunesCode

**Fitness First – Gym Website** Individual Project

- Developed a fitness web application featuring trainer profiles, fitness services, meal plans, pricing information, and an interactive BMI calculator.
- Implemented PHP/MySQL-based user registration and login, database-backed account functionality, and asynchronous communication using JavaScript fetch().
- **Tech:** HTML, CSS, JavaScript, PHP, MySQL
  **GitHub:** github.com/Sajini3655/fitnessfirst
  **Live Demo:** fitnessfirst-sample.page.gd/index.html

## Skills

- **Programming:** Java, Python, JavaScript, C++, TypeScript
- **Web & Backend:** Spring Boot, Spring Security, React, Next.js, Node.js, PHP, REST APIs, WebSockets, STOMP, HTML, CSS, Tailwind CSS
- **Databases & ORM:** PostgreSQL, MySQL, MongoDB, Redis, SQL, Prisma, JPA/Hibernate
- **Software Engineering:** Object-Oriented Programming, Data Structures & Algorithms, Software Testing, Vitest, Playwright, Authentication, Agile/Scrum
- **Distributed Systems & Infrastructure:** RabbitMQ, Docker, Kubernetes, GitHub Actions, CI/CD, Asynchronous Processing, Observability, Prometheus, Grafana, WSO2 API Manager
- **Tools & Development:** Git, GitHub, VS Code, Vercel, Render
- **AI & Data:** Pandas, NumPy, Scikit-learn, Machine Learning Fundamentals, AI API Integration, FastAPI, Groq API
- **Additional Technologies:** OpenGL, GLSL, WebGPU, WGSL, Arduino, ESP32, Embedded Systems

## Certifications & Training

- **WSO2 API Manager 4.7 Developer Fundamentals** – WSO2 (Completed Training)
- **The Complete Full-Stack Web Development Bootcamp** – Udemy (Dec 2025)
- **Career Essentials in Software Development** – Microsoft & LinkedIn (Feb 2025)
- **Python for Beginners** – University of Moratuwa (Mar 2024)

## Leadership & Extracurricular Activities

- **Editorial Head** – IEEE Computer Society
- **Web & Technology Pillar Member** – MoraSpirit
- **Editorial Board Member** – Leo Club of University of Moratuwa
- **Event Coordinator** – Leo Club of University of Moratuwa
- **Editorial Volunteer** – Rotaract Club

## Referees

**Mr. Chirath Perera** — Senior Software Engineer, Creative Software
Phone: +94 76 576 3993

**Ms. Nipuni Premadasa** — Lecturer, University of Moratuwa
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